Tiago Ferrão-Lindqvist

Mechanical Engineer | Python, SQL and Test-Data Automation

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PROFESSIONAL SUMMARY

Mechanical engineer with 12 years of experience in product design and test and validation, transitioning into data engineering. Builds Python tools for scheduled data processing, specification checks and reporting, with SQL Server and Power BI experience. Developed an internal tool used by four other engineers that reduced report preparation from about three days to half a day per test. Fluent English and native Portuguese.

TECHNICAL SKILLS

- Python: pandas, NumPy, Matplotlib, npTDMS; file parsing, data validation and reporting automation.
- SQL and BI: SQL Server queries for trend analysis, writes to an IT-created staging table, Power BI dashboards; SQLite in a personal project.
- Scheduling: Windows Task Scheduler for nightly processing on a lab PC.
- Personal-project tools: Flask, Git; introductory Docker Compose use following a tutorial.
- Engineering: MATLAB/Simulink, LabVIEW, ANSYS Mechanical, SolidWorks, design of experiments, 8D root-cause analysis.

PROFESSIONAL EXPERIENCE

Test & Validation Engineer | Corvane Automotive Components

Setúbal | Mar 2019 - Present

- Developed Python scripts to parse TDMS and CSV output from six test rigs, processing about 2 GB per week, flagging channels outside test specifications and generating standard report figures.
- Scheduled nightly processing through Windows Task Scheduler on a lab PC and wrote summary results to an SQL Server staging table created by IT; the internal tool is used by four other test engineers.
- Reduced report preparation from about three days to half a day per test through automated processing and figure generation.
- Query the plant's SQL Server test-results database for trends across three product generations.
- Built a Power BI dashboard showing pass/fail trends and cycles to failure per product generation, used by about 15 validation and design engineers.
- Plan and run durability, vibration and climatic tests; prepare reports for OEM customers and maintain MATLAB/Simulink models for rig load-profile selection.
- Lead 8D root-cause analysis and correlate test results with FEA; a proposed adjuster rail bracket redesign reduced warranty returns for that part by 40%.
- Trained three apprentices in rig safety and test setup between 2022 and 2024.

Mechanical Design Engineer | Halvik Pumps

Porto | Sep 2014 - Feb 2019

- Designed pump casings and impellers in SolidWorks, produced drawings with GD&T and owned DFMEA for two product lines.

- Wrote a LabVIEW acquisition program for the performance test bench that reduced setup time from about two hours to 30 minutes.
- Performed structural FEA in ANSYS; a casing-family material change saved 12% in material cost.

PERSONAL PROJECT

Bike-sharing Trips API | 2025

- Built a small Flask API over a public bike-sharing dataset loaded into SQLite, with three read-only endpoints; runs locally on a laptop.
- Used Git for the personal repository and followed a tutorial to run the application with Docker Compose.

EDUCATION AND TRAINING

MSc Mechanical Engineering | Sado Valley Polytechnic | 2014

Thesis: Fatigue life prediction of welded brackets under variable-amplitude loading

Online data engineering course | In progress: six of nine modules completed; no certificate yet.

LANGUAGES

Portuguese: native | English: fluent | Spanish: good